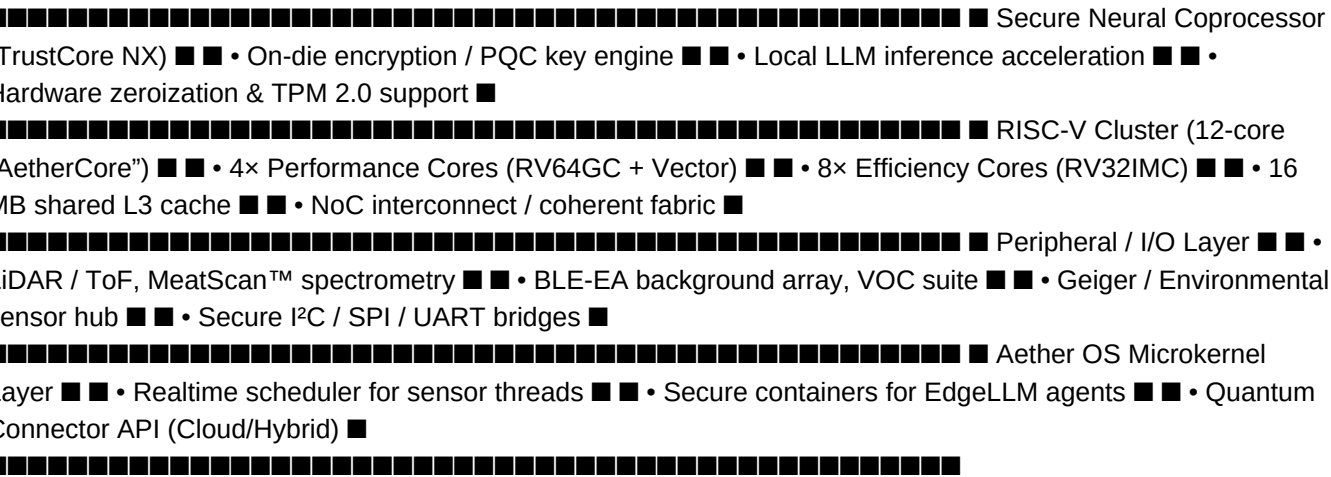


Aether One™ – Custom AI-Accelerated RISC-V 12-Core SoC with Secure Neural Coprocessor

1. Overview

Aether One™ integrates a custom 12-core RISC-V processor cluster with an embedded Secure Neural Coprocessor, designed for field-grade AI inference, sensor fusion, and post-quantum-secure edge computation. The SoC is optimized for modular field devices, combining high performance with low-power operation and tamper-resistant architecture.

2. System Architecture (concept)



3. Key Specifications

Category	Specification
CPU Cluster	12 Cores RISC-V (4xRV64GC + 8xRV32IMC), up to 2.6 GHz
Neural Coprocessor	TrustCore NX, INT8/FP16 matrix engine, 32 TOPS @ 2 W
Security	On-die TPM 2.0, PQC support (NTRU, Kyber), AES-512 engine
Memory Hierarchy	L1: 128 KB/core, L2: 2 MB shared, L3: 16 MB shared
Fabric & Bus	Coherent NoC, PCIe Gen4 x4, LPDDR5-6400 interface
I/O Subsystem	Dual MIPI CSI-2, USB-C 3.2, GPIO, UART, I²C, CAN FD
AI/ML Acceleration	32 TOPS neural unit + vector extensions (RVV 1.0)
Security Monitor	Tamper detection mesh + voltage glitch sensor
Power Envelope	3.5–6 W typical @ 1.2 V core, < 9 W peak
Process Node	7 nm FinFET, TSMC class
Prototype Platform	Raspberry Pi 5 stack with BLE, LiDAR, VOC, Geiger
Operating System	Aether OS (secure microkernel RTOS) + Debian overlay
Language Support	RISC-V GCC / LLVM, Rust, Python, C++ 17

4. Special Features

- Secure Neural Fabric – distributed AI memory with enclave execution.
- Quantum Connector API – bridge to Debian Quantum AI framework.
- Sensor Fusion Bus – unified timestamp domain across LiDAR, BLE, and spectral modules.
- Field-Ready Design – radiation-tolerant layout, -40 to +85 °C range.

5. Development Path

Prototype validation runs on Raspberry Pi 5 using the aetherone_pi5_stack with full sensor suite, simulated coprocessor layer, and secure MQTT/REST/WS interfaces. ASIC implementation to follow, targeting 2026 engineering sample under Aether One initiative.